# ARKFALL — MVP Scope

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The MVP keeps only what is needed to play from start to win or loss; everything else is added in phases afterwards. Internal planning doc, not for submission; the full design is in the ARKFALL GDD.

## MVP: build this first

**Core loop:** place mines → Miners extract ore → ore is processed → Engineers build → call rockets for more colonists → refine Aurite → build the Haven Beacon.

| Area | In the MVP | Simplified compared to the GDD |
|---|---|---|
| Map and camera | One map; pan, rotate, zoom | Iron and copper deposits visible from the start (no scouting). The base has no roof, so no roof-fade effect |
| Colonists | Autonomous, each with a job | Only Hunger, Thirst and Oxygen: at 20, stop and refill; at 0, die. Jobs assigned in the colonist popup |
| Jobs | Engineer, Miner, Farmer | Colonists go to the nearest free job (no priorities) |
| Interior buildings | Exit Door, Oxygenator, Storage, Greenhouse, Iron Processor, Copper Processor, Aurite Processor | Greenhouse grows only vegetables |
| Exterior buildings | Iron Mine, Copper Mine, Water Extractor, Solar Panel, Landing Pad, Haven Beacon | Solar gives constant power (no day/night) |
| Utilities | Electricity, water and oxygen as global numbers (produced vs used) | No grid or pipes to place; oxygen = Oxygenator capacity + mines + suit tank |
| Ore | Mines need Miners and power | Ore goes straight into Storage |
| Rocket | Click the Landing Pad → 10 colonists arrive | No landing animation at first |
| Win / lose | Haven Beacon built (300 Aurite + 100 colonists) / everyone dead | "100 colonists" instead of "100 healthy colonists" until Health exists |

| Area | In the MVP | Simplified compared to the GDD |
|---|---|---|
| UI | Build menu, resource bar, building popup, colonist popup with need bars | — |

That is 13 buildings, 3 jobs and 3 needs, down from 19 buildings, 4 jobs plus a robot, and 6 needs in the full GDD.

## Extras, in a suggested order

**Phase 2: colonist depth**

- [ ] Rest + Sleeping Room
- [ ] Happiness + Arcade (affects Rest)
- [ ] Health states (Healthy / Injured / Dead), Hospital, Medic job, medicinal plants
- [ ] Greenhouse planting-slot panel

**Phase 3: survival pressure**

- [ ] Dust storms (injuries, less solar power, mines clog)
- [ ] Day/night cycle + Energy Battery
- [ ] Wind Turbine
- [ ] Engineers repairing mines and cleaning panels

**Phase 4: polish and logistics**

- [ ] Ore carried in by hand through the Exit Door
- [ ] Building priority system in popups
- [ ] Scout Robot + Charge Pad, hidden deposits, finite deposits
- [ ] Roof fade when zooming over the base
- [ ] Lore intro / story presentation

## Scope notes

- **Phases 2 and 3 can swap.** If the teacher values "survival" most, do dust storms before Health; they are the main source of danger in the full design.
- **Riskiest part of the MVP:** colonist pathfinding plus the job state machine. Get one colonist walking between a mine, Storage and the Exit Door before building anything else.